

Nibi

Availability: since 31 July 2025
SSH login node: <code>nibi.alliancecan.ca</code>
Automation node: <code>robot.nibi.alliancecan.ca</code>
Web interface: <code>ondemand.sharcnet.ca</code> (https://ondemand.sharcnet.ca)
Globus collection: <code>alliancecan#nibi</code> (https://app.globus.org/file-manager?origin_id=07baf15f-d7fd-4b6a-bf8a-5b5ef2e229d3)
Data transfer node (rsync, scp, sftp,...): use login nodes
Portal: <code>portal.nibi.sharcnet.ca</code> (https://portal.nibi.sharcnet.ca)
AI Platform: <code>brine.sharcnet.ca/ui/</code>

Nibi, the Anishinaabemowin word for water, is a general purpose cluster of 134,400 CPU cores and 288 H100 NVIDIA GPUs. Built by **Hypertec** (<https://www.hypertec.com/>), the cluster is hosted and operated by **SHARCNET** (<https://www.sharcnet.ca/>) at University of Waterloo.

Contents

Access

Storage

Interconnect fabric

Node characteristics

GPU instances

Site specifics

Internet access

/project and /nearline spaces

/scratch quota

Access through Open OnDemand (OOD)

Use of JupyterLab via OOD

Support for VDI via OOD

AMD MI300A nodes

Oops, I accidentally deleted my files, what should I do?

Brine: AI-as-a-Service Platform

Receiving Access

LiteLLM Interface

Service Status

Access

Each researcher must **request access in CCDB (https://ccdb.alliancecan.ca/me/access_systems)**, via Resources--> Access Systems.

- Select Nibi from the list on the left.
- Select I request access.

It can take up to one hour for your access to be enabled.

Storage

Parallel storage: 25 petabytes, all **SSD (https://en.wikipedia.org/wiki/Solid-state_drive)** from **VAST Data (<https://www.vastdata.com/>)** for /home, /project and /scratch.

Note that Vast implements space accounting for quotas differently. You are "charged" for the apparent size of your files. This is in contrast to some Lustre configurations, which transparently compress files and charge for the space used after compression.

Note also that Nibi is using a new, experimental mechanism for handling /scratch. As on all systems, you have a soft and a hard limit, but on Nibi, the soft limit is low (1TB), and you have a 60d grace period. After the grace period expires, the soft limit is enforced (no further file creation/expansion). To rectify this, your usage must drop below the soft limit.

Interconnect fabric

- Nokia 200/400G ethernet
 - 200 Gbit/s network bandwidth for CPU nodes,
 - 200 Gbit/s non-blocking network bandwidth between all Nvidia GPU nodes,
 - 200 Gbit/s network bandwidth between all AMD GPU nodes,
 - 24x100 Gbit/s connection to the VAST storage nodes,
 - 2:1 blocking at 400 Gbit/s uplinks for all compute nodes.

The topology of the network is described in the file

`/etc/slurm/topology.conf`

For better performance of tightly coupled multinode jobs, you may constrain them to use only one network switch, by adding this next option to your job submission script

```
#SBATCH --switches=1
```

Node characteristics

nodes	cores	available memory	node-local storage	CPU	GPU
700	192	748G or 766000M	3T	2 x Intel 6972P @ 2.4 GHz, 384MB cache L3	
10	192	6000G or 6144000M	3T	2 x Intel 6972P @ 2.4 GHz, 384MB cache L3	
36	112	2000G or 2048000M	11T	2 x Intel 8570 @ 2.1 GHz, 300MB cache L3	8 x Nvidia H100 SXM (80 GB), connected via NVLink
6	96	495G or 507000M	3T	4 x AMD MI300A @ 2.1GHz (Zen4+CDNA3)	The CPU cores and CDNA3-based GPUs are in the same socket and share a unified memory. See section below for use instructions.

GPU instances

Available GPU instance names are:

Model or instance		Short name	Without unit	By memory	Full name
GPU	H100-80gb	h100	h100	h100_80gb	nvidia_h100_80gb_hbm3
MIG	H100-1g.10gb	h100_1g.10gb	h100_1.10	h100_10gb	nvidia_h100_80gb_hbm3_1g.10gb
	H100-2g.20gb	h100_2g.20gb	h100_2.20	h100_20gb	nvidia_h100_80gb_hbm3_2g.20gb
	H100-3g.40gb	h100_3g.40gb	h100_3.40	h100_40gb	nvidia_h100_80gb_hbm3_3g.40gb

To request one or more full H100 GPUs, you need to use one of the following Slurm options:

- **One H100-80gb** : `--gpus=h100:1` or `--gpus=h100_80gb:1`
- **Multiple H100-80gb** per node :
 - `--gpus-per-node=h100:2`
 - `--gpus-per-node=h100:3`
 - `--gpus-per-node=h100:4`

- **For multiple full H100 GPUs** spread anywhere: `--gpus=h100:n` (replace n with the number of GPUs you want)

Approximately half of the GPU nodes are configured with **MIG technology**. Only 3 GPU instance sizes are available:

- **H100-1g.10gb**: 1/8th of the computing power with 10GB GPU memory
- **H100-2g.20gb**: 2/8th of the computing power with 20GB GPU memory
- **H100-3g.40gb**: 3/8th of the computing power with 40GB GPU memory

To request **one and only one GPU instance** for your compute job, use the corresponding option:

- **H100-1g.10gb** : `--gpus=h100_1g.10gb:1`
- **H100-2g.20gb** : `--gpus=h100_2g.20gb:1`
- **H100-3g.40gb** : `--gpus=h100_3g.40gb:1`

The maximum recommended number of CPU cores and system memory per GPU instance is listed **in this table**.

Site specifics

Internet access

All nodes on Nibi have internet access, no special firewall permission or proxying is necessary.

/project and /nearline spaces

User directories are no longer created by default in /project or /nearline. Users can always create their own directories in the group's /project or /nearline using `mkdir`. This allows groups to decide how their /project or /nearline spaces are organized for sharing data amongst group members. **Index files** and other small files are not archived to tape or backed up on Nibi /nearline.

/scratch quota

An 1 TB soft quota on /scratch applies to each user. This soft quota can be exceeded for up to 60 days after which no additional files may be written to /scratch. Files may be written again once the user has removed or deleted enough files to bring their total /scratch use under 1 TB. See **Storage and file management** for more information.

Access through Open OnDemand (OOD)

You can now access the Nibi cluster simply through a web browser. Nibi uses Open OnDemand (OOD), a web-based platform that simplifies cluster access by providing a web interface to the login nodes and a remote desktop environment. To log into Nibi, go to **<https://ondemand.sharcnet.ca/>**

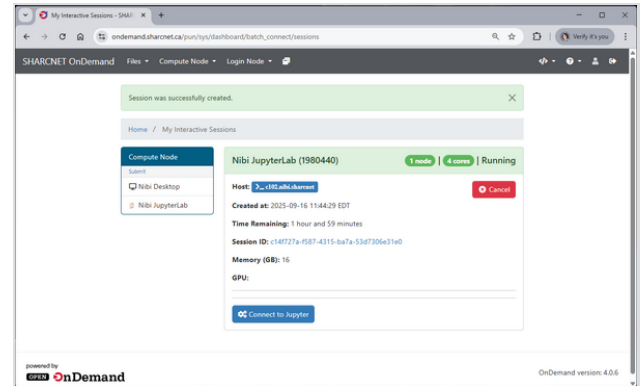
and sign in with **multifactor authentication**; you will see a user-friendly interface offering options to open a Bash shell terminal or launch a remote desktop session.

Use of JupyterLab via OOD

You can run JupyterLab interactively via the Nibi Open OnDemand **portal** (<https://ondemand.sharcnet.ca>).

Option 1: Working with a pre-configured environment (same as from **JupyterHub**)

After logging into the Nibi Open OnDemand **portal** (<https://ondemand.sharcnet.ca>), click on *Compute Node* from the top menu and select *Nibi JupyterLab*. This will open a page with a form where you can request a new Nibi JupyterLab session.



After completing the form with your requirement details, click on *Launch* to submit your request. Once the status of the requested Nibi JupyterLab changes to *Running*, click on *Connect to Jupyter* to open JupyterLab in your web browser.

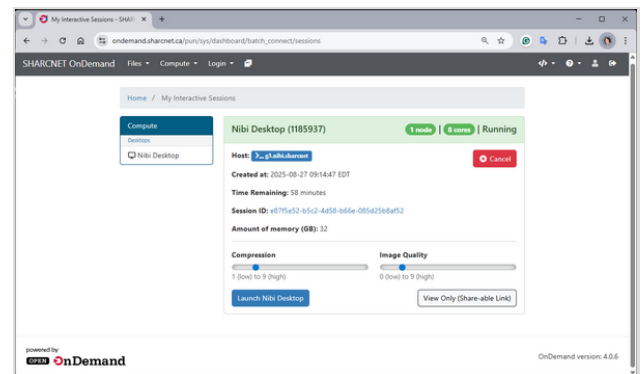
More details about the pre-configured JupyterLab are **described here**.

Option 2: Working with a self-built **Python virtual environment**

After logging into the Nibi Open OnDemand **portal** (<https://ondemand.sharcnet.ca>), click on *Compute Node* from the top menu and select *Compute Desktop*. This will open a page with a form where you can request a new Compute Desktop session.

After completing the form with your requirement details, click on *Launch* to submit your request. Once the status of the requested Compute Desktop changes to *Running*, click on *Launch Compute Desktop* to connect to the desktop. A Linux desktop will appear.

On the Compute Desktop, right-click the mouse in any blank area to display a shortcut menu. Select *Open in Terminal* to open a terminal window, where you can create or activate your Python virtual environment that has JupyterLab installed.



If you do not have JupyterLab installed in the Python virtual environment you would like to work with, you can have it installed with the command

```
(your_python_ENV) [username@<node>.nibi]$ pip install --no-index jupyterlab
```

Then, you can launch JupyterLab from your Python virtual environment with the command

```
(your_python_ENV) [username@<node>.nibi]$ jupyter-lab --notebook-dir $HOME
```

You will see JupyterLab is opened in the web browser on the desktop with your \$HOME contents listed in the file browser panel on JupyterLab.

Support for VDI via OOD

Nibi no longer offers Virtual Desktop Infrastructure (VDI). Instead, it provides a remote desktop environment through the **portal (<https://ondemand.sharcnet.ca/>)** of Open OnDemand (OOD), offering improved hardware performance and software support.

AMD MI300A nodes

At this time, the MI300A should be scheduled as full nodes. It is your responsibility to make sure the processes inside the job run with the correct core and memory bindings. Here is a representative job script that uses 4 processes.

NOTE: Your code must be compiled with ROCm to use the MI300A nodes properly. Code compiled with CUDA will not work, as it will not be able to use the AMD GPUs.

As of May 2026, work has just started to support the MI300A in our software stack and there are no modules available with ROCm support. You can install software yourself, building against the ROCm toolkit we installed in the /opt/rocm directory. Please write to **technical support** if you run into problems.

File : simple_job.sh

```
#!/bin/bash
#SBATCH --account=def-someuser
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --cpus-per-task=24
#SBATCH --gpus=mi300a:4
#SBATCH --mem=100g
#SBATCH --time=00:02:00
```

```
# verify GPUS are available (optional)

rocm-smi

# run program compiled with ROCm support for MI300A
```

Oops, I accidentally deleted my files, what should I do?

A backup mechanism on Nibi takes a snapshot of your files on /home and /project every 30 minutes, and saves the snapshots for two weeks. If you accidentally delete a file, you may be able to retrieve it from these snapshots, providing the file was deleted less than two weeks back. However, if you make changes to a file after the most recent snapshot and then delete it, the changes cannot be recovered.

To find a deleted file, use the `oops` command to check the current directory, or give an optional directory name to check there instead. To recover a file, copy it from the path returned by `oops` using standard tools like `cp`. Snapshots are read-only; you cannot delete or change files in snapshots, you must copy them first. Do not refer to files in snapshots in your job scripts.

```
[username@<node>.nibi]$ ls
dont_delete_me.txt
[username@<node>.nibi]$ rm dont_delete_me.txt
[username@<node>.nibi]$ ls
[username@<node>.nibi]$ oops
Deleted files found in snapshots:
./snapshot/backup_2026-04-01_18_00_00_UTC/dont_delete_me.txt
Files created less than 9 min ago (2026-04-01 14:00:00-04:00) are not yet backed up
Files deleted more than 0 days ago (2026-04-01 13:30:00-04:00) please submit a help ticket
[username@<node>.nibi]$ cp ./snapshot/backup_2026-04-01_18_00_00_UTC/dont_delete_me.txt .
[username@<node>.nibi]$ ls
dont_delete_me.txt
```

Brine: AI-as-a-Service Platform

As part of a new effort, SHARCNET has launched an AI-as-a-Service platform hosted from within the Nibi datacenter. This service gives Canadian researchers access to hosted AI models on SHARCNET infrastructure, without the usual HPC queue wait.

Brine provides an always-on, OpenAI-compatible API for hosted models that support chat, tool calling, and audio transcription. Model availability and parameters are outlined **here (<https://github.com/sharcnet/brine-examples/tree/main/model-cards>)**.

Brine runs on hardware in the Graham data centre, so data is processed in Canada rather than sent to a commercial provider outside the country. Even so, users should not submit sensitive data. This includes personal or health information, student or financial records, credentials, confidential third-

party data, export-controlled data, Indigenous or community-governed data, and anything you are not authorized to send to a centrally hosted AI service.

To request access, email help@sharcnet.ca and mention SHARCNET Brine.

Receiving Access

After accepting the terms of service, you will receive an email invitation to your mailbox prompting you to register for a new account with LiteLLM. This is a new account for only this service.

Note: We have received alerts that some mailboxes report the invitation as spam. Please check your junk mail.

Upon logging in, you should be presented with an interface that prompts you to create a new key. Create the key for yourself, name it, and do not share it with others.

This key is what is required to access the service as an API outlined in the **brine examples (<https://github.com/sharcnet/brine-examples>)** repository. This includes Python, Javascript, and Curl examples.

LiteLLM Interface

The back end of this service is powered by **LiteLLM (<https://www.litellm.ai/>)** and offers several interesting features outside of API key creation and model service.

- Monitor your own token usage and key activity
- Compare the outputs of models in the playground feature
- Add Vector stores
- Manage MCP connections

Service Status

To keep up to date with changes to this service, an invitation to a slack channel is available upon request.

For outage alerts and updates, please see the status page **here (<https://status.alliancecan.ca/system/Nibi%20AIaaS:%20Brine>)**.

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This page was last edited on 1 September 2026, at 16:28.